MASERs and Quantum Heat Engines

PH4104 Term Project

Sabarno Saha(22MS037) Diptanuj Sarkar(22MS038) Debayan Sarkar(22MS002)

April 17, 2026

Table of Contents

- Introduction
- MASERs
- Quantum Heat Engines
- Scovil Schulz-DuBois(SSD) Model
- Modern Work
- Conclusion

Introduction

- MASER: Microwave Amplification by Stimulated Emission of Radiation.

Introduction

- MASER: Microwave Amplification by Stimulated Emission of Radiation.
- Quantum Heat Engine: A device that converts heat into work using quantum systems.

Introduction

- MASER: Microwave Amplification by Stimulated Emission of Radiation.
- Quantum Heat Engine: A device that converts heat into work using quantum systems.
- Scovil Schulz-DuBois(SSD) Model: The original quantum heat engine done with respect to MASERs.

Introduction

- MASER: Microwave Amplification by Stimulated Emission of Radiation.
- Quantum Heat Engine: A device that converts heat into work using quantum systems.
- Scovil Schulz-DuBois(SSD) Model: The original quantum heat engine done with respect to MASERs.

The Three State MASER served as the first example of a Quantum Heat Engine, as proposed by Scovil and Schulz-DuBois in 1959. This launched the field of Quantum Thermodynamics, which studies the thermodynamic properties of quantum systems and their applications in quantum technologies.

Table of Contents

2 MASERs

- Introduction
- MASERs
- Quantum Heat Engines
- Scovil Schulz-DuBois(SSD) Model
- Modern Work
- Conclusion

The Three-Level MASER

Let us define three energy levels: $E_1 < E_2 < E_3$. The total number of atoms in a particular state is given by N_i, where i is the index for that level. Note that we fix the number of atoms available as: $N = N_1 + N_2 + N_3$

There are two distinct types of transitions:

- **Stimulated Transitions** (denoted by W here): These are stimulated by external electromagnetic fields. W_p is the pumping rate between the levels 1 and 3. Since stimulated emission and absorption probabilities are symmetric, we have $W_{13} = W_{31} = W_p$.
- **Intrinsic Relaxation Rates** (denoted here with Γ): These are the transition in the system that aim to restore thermal equilibrium. In a solid-state system at microwave frequencies, spontaneous photon emission (Given by the Einstein A coefficient, which scales as ω^3, ω being the frequency associated with the transition) is for all practical purposes nearly zero. Therefore, these Γ_{ij} relaxation process are almost entirely non-radiative in character.

Rate Equations

2 MASERs

Rate Equations

$$\frac{dN_3}{dt} = W_p(N_1 - N_3) - (\Gamma_{32} + \Gamma_{31})N_3 + \Gamma_{23}N_2 + \Gamma_{13}N_1 \tag{1}$$

$$\frac{dN_2}{dt} = \Gamma_{32}N_3 - (\Gamma_{23} + \Gamma_{21})N_2 + \Gamma_{12}N_1 \tag{2}$$

$$\frac{dN_1}{dt} = W_p(N_3 - N_1) + \Gamma_{31}N_3 + \Gamma_{21}N_2 - (\Gamma_{13} + \Gamma_{12})N_1 \tag{3}$$

where N_i is the population of the i-th energy level. It also has the implicit constraint that $N = N_1 + N_2 + N_3$, where N is the total number of atoms in the system.

Steady-State and Pump Saturation

2 MASERs

Let us evaluate the system at steady-state (i.e. assume $\frac{dN_i}{dt} = 0 \forall i$).
Now, we apply a **strong** pump field - this means that the stimulated pump rate is much greater than the natural relaxation rates ($W_p \gg \Gamma_{ij}$). Notice that the only way that the N_3 rate equation would ever be balanced under both of these assumptions would be if $N_1 \approx N_3$ and thus the first term on the RHS vanishes (and takes along the large W_p factor). Let us define $N_1 \approx N_3 \equiv n$.
Substituting these conditions into the rate equation for N_2, we get:

$$0 = \Gamma_{32} n - (\Gamma_{23} + \Gamma_{21}) N_2 + \Gamma_{12} n$$

Rearranging,

$$N_2 = n \frac{\Gamma_{32} + \Gamma_{12}}{\Gamma_{21} + \Gamma_{23}}$$

Population Inversion

2 MASERs

The aim is to amplify a signal across the $E_2 \to E_1$ transition in this system. The necessary condition for this stimulated emission to outpace absorption is $\Delta N = N_2 - N_1 > 0$ (the usual population inversion condition). Plugging in the expressions derived above,

$$\Delta N = N_2 - n = n\left[\frac{\Gamma_{23} + \Gamma_{12}}{\Gamma_{21} + \Gamma_{23}} - 1\right]$$

This simplifies to:

$$\Delta N = n\left[\frac{\Gamma_{32} + \Gamma_{12} - \Gamma_{21} - \Gamma_{23}}{\Gamma_{21} + \Gamma_{23}}\right]$$

For the $\Delta N > 0$ condition to be satified, we must have:

$$\Gamma_{32} + \Gamma_{12} > \Gamma_{21} + \Gamma_{23}$$

Physical Motivations

2 MASERs

- The LHS of the above inequality represents the total rate of transitions that populate the E_2 level, from both the E_3 and E_1 levels. The RHS represents the total rate of transitions that depopulate the E_2 level, to both the E_3 and E_1 levels.

Physical Motivations

2 MASERs

- The LHS of the above inequality represents the total rate of transitions that populate the E_2 level, from both the E_3 and E_1 levels. The RHS represents the total rate of transitions that depopulate the E_2 level, to both the E_3 and E_1 levels.
- Therefore, the condition for population inversion can be interpreted as the requirement that the total rate of transitions populating the E_2 level must be greater than the total rate of transitions depopulating the E_2 level.

Physical Motivations

2 MASERs

- The LHS of the above inequality represents the total rate of transitions that populate the E_2 level, from both the E_3 and E_1 levels. The RHS represents the total rate of transitions that depopulate the E_2 level, to both the E_3 and E_1 levels.
- Therefore, the condition for population inversion can be interpreted as the requirement that the total rate of transitions populating the E_2 level must be greater than the total rate of transitions depopulating the E_2 level.
- In modern implementations of the 3 state Laser, Γ_{32} is quite large and Γ_{23} is quite small, because the $|2\rangle$ level is typically a metastable state. This means that the condition for population inversion is easily satisfied.

MASERs vs LASERs

2 MASERs

- Both MASERs and LASERs operate on the principle of stimulated emission, where an incoming photon stimulates an excited atom to emit a photon of the same frequency, phase, and direction.

MASERs vs LASERs

- Both MASERs and LASERs operate on the principle of stimulated emission, where an incoming photon stimulates an excited atom to emit a photon of the same frequency, phase, and direction.
- The main difference between MASERs and LASERs is the frequency of the emitted radiation. MASERs emit microwave radiation, while LASERs emit visible or infrared light.

MASERs vs LASERs

2 MASERs

- Both MASERs and LASERs operate on the principle of stimulated emission, where an incoming photon stimulates an excited atom to emit a photon of the same frequency, phase, and direction.
- The main difference between MASERs and LASERs is the frequency of the emitted radiation. MASERs emit microwave radiation, while LASERs emit visible or infrared light.
- The Energy gaps in the MASER system are much smaller than those in the LASER system, which is why thermal fluctuations cause significant problems in MASERs and need to be cryogenically cooled to near absolute zero.

MASERs vs LASERs

2 MASERs

- Both MASERs and LASERs operate on the principle of stimulated emission, where an incoming photon stimulates an excited atom to emit a photon of the same frequency, phase, and direction.
- The main difference between MASERs and LASERs is the frequency of the emitted radiation. MASERs emit microwave radiation, while LASERs emit visible or infrared light.
- The Energy gaps in the MASER system are much smaller than those in the LASER system, which is why thermal fluctuations cause significant problems in MASERs and need to be cryogenically cooled to near absolute zero.

Stimulated Emission

- Stimulated Emission is an interesting phenomena, where a system decays from an energetic state to a less energetic state due to an external photon coupling to it. The crucial phenomena is that the photon radiated from the system due to this event is **identical to the original external photon** in all ways.

Stimulated Emission

- Stimulated Emission is an interesting phenomena, where a system decays from an energetic state to a less energetic state due to an external photon coupling to it. The crucial phenomena is that the photon radiated from the system due to this event is **identical to the original external photon** in all ways.
- We present the semiclassical derivation for a two-level system, with energy levels E_1 and E_2, where $E_2 > E_1$.
- Consider the free Hamiltonian of the system to be H_0, with the eigensystem $|1\rangle, E_1$ and $|2\rangle, E_2$. This atom is then hit with a monochromatic linearly polarized electromagnetic wave, which we treat classically. We assume that the wavelength of the EM wave is much larger than the size of the atom, so we can ignore spatial variations in the field across the atom. This is called the **Dipole Approximation**.

Stimulated Emission

2 MASERs

- Stimulated Emission is an interesting phenomena, where a system decays from an energetic state to a less energetic state due to an external photon coupling to it. The crucial phenomena is that the photon radiated from the system due to this event is **identical to the original external photon** in all ways.
- We present the semiclassical derivation for a two-level system, with energy levels E_1 and E_2, where $E_2 > E_1$.
- Consider the free Hamiltonian of the system to be H_0, with the eigensystem $|1\rangle, E_1$ and $|2\rangle, E_2$. This atom is then hit with a monochromatic linearly polarized electromagnetic wave, which we treat classically. We assume that the wavelength of the EM wave is much larger than the size of the atom, so we can ignore spatial variations in the field across the atom. This is called the **Dipole Approximation**.
- The interaction Hamiltonian between the atom and the EM wave can be written as:

$$H_I = -\mathbf{d} \cdot \mathbf{E}(t) = -\mathbf{d} \cdot \mathbf{E}_0 \cos(\omega t)$$

Stimulated Emission

- The dipole interaction creates off-diagonal terms in the Hamiltonian, which can cause transitions between the two energy levels. The transition probability from $|2\rangle$ to $|1\rangle$ due to the interaction with the EM wave can be calculated using time-dependent perturbation theory.

Stimulated Emission

- The dipole interaction creates off-diagonal terms in the Hamiltonian, which can cause transitions between the two energy levels. The transition probability from $|2\rangle$ to $|1\rangle$ due to the interaction with the EM wave can be calculated using time-dependent perturbation theory.
- Time dependent perturbation theory tells us that the transition probability is proportional to two oscillatory terms: one with frequency $\omega_{21} + \omega$ and one with frequency $\omega_{21} - \omega$, where $\omega_{21} = (E_2 - E_1)/\hbar$ is the natural transition frequency of the atom.

Stimulated Emission

2 MASERs

- The dipole interaction creates off-diagonal terms in the Hamiltonian, which can cause transitions between the two energy levels. The transition probability from $|2\rangle$ to $|1\rangle$ due to the interaction with the EM wave can be calculated using time-dependent perturbation theory.
- Time dependent perturbation theory tells us that the transition probability is proportional to two oscillatory terms: one with frequency $\omega_{21} + \omega$ and one with frequency $\omega_{21} - \omega$, where $\omega_{21} = (E_2 - E_1)/\hbar$ is the natural transition frequency of the atom.
- Assuming that the EM wave is near-resonant with the atomic transition (i.e. $\omega \approx \omega_{21}$), the fast oscillating term with frequency $\omega_{21} + \omega$ can be integrated out over a small time interval, leaving us with the term that oscillates with frequency $\omega_{21} - \omega$. This is called the **Rotating Wave Approximation**.

Stimulated Emission

2 MASERs

- The dipole interaction creates off-diagonal terms in the Hamiltonian, which can cause transitions between the two energy levels. The transition probability from $|2\rangle$ to $|1\rangle$ due to the interaction with the EM wave can be calculated using time-dependent perturbation theory.
- Time dependent perturbation theory tells us that the transition probability is proportional to two oscillatory terms: one with frequency $\omega_{21} + \omega$ and one with frequency $\omega_{21} - \omega$, where $\omega_{21} = (E_2 - E_1)/\hbar$ is the natural transition frequency of the atom.
- Assuming that the EM wave is near-resonant with the atomic transition (i.e. $\omega \approx \omega_{21}$), the fast oscillating term with frequency $\omega_{21} + \omega$ can be integrated out over a small time interval, leaving us with the term that oscillates with frequency $\omega_{21} - \omega$. This is called the **Rotating Wave Approximation**.
- Using these two approximations, we find that the transition probability from $|2\rangle$ to $|1\rangle$ is proportional to the intensity of the EM wave, which is a hallmark of stimulated emission.

Table of Contents

3 Quantum Heat Engines

- Introduction
- MASERs
- Quantum Heat Engines
- Scovil Schulz-DuBois(SSD) Model
- Modern Work
- Conclusion

Energy Levels

- One crucial difference between classical and quantum heat engines is that in a quantum heat engine, the working substance is a quantum system with discrete energy levels, rather than a classical system with continuously varying energy.

Energy Levels

- One crucial difference between classical and quantum heat engines is that in a quantum heat engine, the working substance is a quantum system with discrete energy levels, rather than a classical system with continuously varying energy.
- This means that the thermodynamic processes in a quantum heat engine involve transitions between discrete energy levels. This is similar to the transitions between energy levels in a MASER, which is why the SSD model is a natural example of a quantum heat engine.

Energy Levels

- One crucial difference between classical and quantum heat engines is that in a quantum heat engine, the working substance is a quantum system with discrete energy levels, rather than a classical system with continuously varying energy.
- This means that the thermodynamic processes in a quantum heat engine involve transitions between discrete energy levels. This is similar to the transitions between energy levels in a MASER, which is why the SSD model is a natural example of a quantum heat engine.
- As a result, fluctuations in the system manifest as transitions between these discrete energy levels.

One can in general Manipulate these energy levels by applying external fields, which forms the basis of work extraction in a quantum heat engine.

Thermal Reservoirs

- In a quantum heat engine, the system is coupled to thermal reservoirs that can exchange energy with the system. These reservoirs are typically modeled as large collections of quantum systems that are in thermal equilibrium at a certain temperature.

Thermal Reservoirs

- In a quantum heat engine, the system is coupled to thermal reservoirs that can exchange energy with the system. These reservoirs are typically modeled as large collections of quantum systems that are in thermal equilibrium at a certain temperature.
- The coupling between the system and the reservoirs allows for the exchange of energy in the form of heat, which is essential for the operation of the heat engine.

Thermal Reservoirs

- In a quantum heat engine, the system is coupled to thermal reservoirs that can exchange energy with the system. These reservoirs are typically modeled as large collections of quantum systems that are in thermal equilibrium at a certain temperature.
- The coupling between the system and the reservoirs allows for the exchange of energy in the form of heat, which is essential for the operation of the heat engine.
- One common example of a thermal reservoir is a large collection of harmonic oscillators at a fixed temperature. This model is often used to describe the environment in which the quantum system is embedded, for example a three level system in an optical cavity coupled to the electromagnetic field modes of the cavity.

Thermal Reservoirs

An example of a system coupled to a thermal reservoir is a two-level atom coupled to the electromagnetic field modes of an optical cavity. This is similar to the MASER system, where the three-level system is coupled to two thermal reservoirs (the hot and cold reservoirs) that allow for the exchange of energy in the form of heat, and the cavity mode allows for the exchange of energy in the form of work.

Density Matrix Formalism

- To properly describe the dynamics of a quantum system that is not in isolation, we need to use the density matrix formalism. This allows us to describe the state of the system as a statistical mixture of pure states, which is necessary when the system is coupled to a thermal reservoir.

Density Matrix Formalism

- To properly describe the dynamics of a quantum system that is not in isolation, we need to use the density matrix formalism. This allows us to describe the state of the system as a statistical mixture of pure states, which is necessary when the system is coupled to a thermal reservoir.
- The density matrix ρ of a quantum system is defined as:

$$\rho = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

where p_i is the probability of the system being in the pure state $\left|\psi_i\right\rangle$.

Density Matrix Formalism

- To properly describe the dynamics of a quantum system that is not in isolation, we need to use the density matrix formalism. This allows us to describe the state of the system as a statistical mixture of pure states, which is necessary when the system is coupled to a thermal reservoir.
- The density matrix ρ of a quantum system is defined as:

$$\rho = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

  where p_i is the probability of the system being in the pure state $\left|\psi_i\right\rangle$.
- The density matrix allows us to calculate expectation values of observables, using the simple formula:

$$\left\langle \hat{O} \right\rangle = \mathrm{Tr}\left(\rho \hat{O}\right)$$

Quantum Heat and Work

3 Quantum Heat Engines

- To properly develop the concept of a quantum heat engine, we need to define what we mean by heat and work in a quantum context.

Quantum Heat and Work

3 Quantum Heat Engines

- To properly develop the concept of a quantum heat engine, we need to define what we mean by heat and work in a quantum context.
- To do that let us assume, that the first law of thermodynamics holds in the quantum regime, which means that the change in internal energy of the system is equal to the heat absorbed from the reservoir minus the work done by the system on its surroundings.

Quantum Heat and Work

3 Quantum Heat Engines

- To properly develop the concept of a quantum heat engine, we need to define what we mean by heat and work in a quantum context.
- To do that let us assume, that the first law of thermodynamics holds in the quantum regime, which means that the change in internal energy of the system is equal to the heat absorbed from the reservoir minus the work done by the system on its surroundings.
- The average energy of the system can be calculated using the density matrix as:

$$\langle E \rangle = \mathrm{Tr}(\rho H)$$

where H is the Hamiltonian of the system.

Quantum Heat and Work

3 Quantum Heat Engines

- To properly develop the concept of a quantum heat engine, we need to define what we mean by heat and work in a quantum context.
- To do that let us assume, that the first law of thermodynamics holds in the quantum regime, which means that the change in internal energy of the system is equal to the heat absorbed from the reservoir minus the work done by the system on its surroundings.
- The average energy of the system can be calculated using the density matrix as:

$$\langle E \rangle = \mathrm{Tr}(\rho H)$$

  where H is the Hamiltonian of the system.
- The change in internal energy can be expressed as:

$$d\langle E \rangle = \mathrm{Tr}(d\rho H) + \mathrm{Tr}(\rho dH)$$

Quantum Heat and Work

3 Quantum Heat Engines

- Thermodynamically Work was done changing volumes using a piston, which means that work was done by changing the Hamiltonian of the system. Therefore, we can identify the work done by the system as:

$$\delta W = -\operatorname{Tr}(\rho dH)$$

Quantum Heat and Work

3 Quantum Heat Engines

- Thermodynamically Work was done changing volumes using a piston, which means that work was done by changing the Hamiltonian of the system. Therefore, we can identify the work done by the system as:

$$\delta W = -\operatorname{Tr}(\rho dH)$$

- Heat, on the other hand, was done by allowing the system to exchange energy with the thermal reservoir, which means that heat was done by changing the state of the system. Therefore, we can identify the heat absorbed by the system as the change in populations of the energy levels:

$$\delta Q = \operatorname{Tr}(d\rho H)$$

Open Quantum Systems

- These definitions of heat and work are quite arbitrary and use the assumption that the first law of thermodynamics holds in the quantum regime. To properly describe the dynamics of a quantum system that is coupled to a thermal reservoir, we need to use the theory of open quantum systems.

Open Quantum Systems

3 Quantum Heat Engines

- These definitions of heat and work are quite arbitrary and use the assumption that the first law of thermodynamics holds in the quantum regime. To properly describe the dynamics of a quantum system that is coupled to a thermal reservoir, we need to use the theory of open quantum systems.
- Similar to the Einstein Rate Equations, the dynamics of the density matrix of an open quantum system can be described by a master equation, which is a differential equation that describes how the density matrix evolves in time.

Open Quantum Systems

3 Quantum Heat Engines

- These definitions of heat and work are quite arbitrary and use the assumption that the first law of thermodynamics holds in the quantum regime. To properly describe the dynamics of a quantum system that is coupled to a thermal reservoir, we need to use the theory of open quantum systems.
- Similar to the Einstein Rate Equations, the dynamics of the density matrix of an open quantum system can be described by a master equation, which is a differential equation that describes how the density matrix evolves in time.
- The most common form of the master equation is the Lindblad Master Equation, which is given by:

$$\frac{\partial \rho_S}{\partial t} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho_S\} \right)$$

where H_S is the Hamiltonian of the system, γ_k are the rates of the different dissipative processes, and L_k are the Lindblad operators that describe the different jumps in the system due to the coupling to the reservoir.

Open Quantum Systems

3 Quantum Heat Engines

- We shall not dwell on this much further, since the Lindblad Master Equation is a quite complex topic and requires a lot of background in quantum mechanics and open quantum systems to properly understand. However, it is important to note that the Lindblad Master Equation provides a rigorous framework for describing the dynamics of open quantum systems, and is essential for understanding the operation of quantum heat engines.
- The Model we are going to describe now can also be formulated in this formalism, which can give us much more insight into the dynamics of the system and the thermodynamic processes involved.

Table of Contents

4 Scovil Schulz-DuBois(SSD) Model

- Introduction
- MASERs
- Quantum Heat Engines
- Scovil Schulz-DuBois(SSD) Model
- Modern Work
- Conclusion

Scovil Schulz-DuBois(SSD) Model

- The SSD model describes a three-level system that can operate as a quantum heat engine.

Scovil Schulz-DuBois(SSD) Model

- The SSD model describes a three-level system that can operate as a quantum heat engine.
- The three levels are denoted as $|1\rangle$, $|2\rangle$, and $|3\rangle$, with energies $E_1 < E_2 < E_3$.

Scovil Schulz-DuBois(SSD) Model

- The SSD model describes a three-level system that can operate as a quantum heat engine.
- The three levels are denoted as $|1\rangle$, $|2\rangle$, and $|3\rangle$, with energies $E_1 < E_2 < E_3$.
- The system is coupled to two thermal reservoirs: a hot reservoir at temperature T_h and a cold reservoir at temperature T_c.

Scovil Schulz-DuBois(SSD) Model

- The SSD model describes a three-level system that can operate as a quantum heat engine.
- The three levels are denoted as $|1\rangle$, $|2\rangle$, and $|3\rangle$, with energies $E_1 < E_2 < E_3$.
- The system is coupled to two thermal reservoirs: a hot reservoir at temperature T_h and a cold reservoir at temperature T_c.
- The engine operates by absorbing heat from the hot reservoir, performing work, and releasing heat to the cold reservoir.

The Model

4 Scovil Schulz-DuBois(SSD) Model

- The characteristic frequencies of the transitions are defined as:

$$\omega_{31} = \omega_h = \frac{E_3 - E_1}{\hbar}$$
$$\omega_{21} = \omega_c = \frac{E_2 - E_1}{\hbar}$$
$$\omega_{32} = \nu = \frac{E_3 - E_2}{\hbar}$$

The Model

4 Scovil Schulz-DuBois(SSD) Model

- The characteristic frequencies of the transitions are defined as:

$$\omega_{31} = \omega_h = \frac{E_3 - E_1}{\hbar}$$
$$\omega_{21} = \omega_c = \frac{E_2 - E_1}{\hbar}$$
$$\omega_{32} = \nu = \frac{E_3 - E_2}{\hbar}$$

- The transition $|3\rangle \leftrightarrow |1\rangle$ is coupled to the hot reservoir, while the transition $|2\rangle \leftrightarrow |1\rangle$ is coupled to the cold reservoir.

The Model

4 Scovil Schulz-DuBois(SSD) Model

- There is a filter that allows only frequencies near the vicinity of ω_h to be absorbed from the hot reservoir and only frequencies near the vicinity of ω_c to be absorbed by the cold reservoir.

The Model

4 Scovil Schulz-DuBois(SSD) Model

- There is a filter that allows only frequencies near the vicinity of ω_h to be absorbed from the hot reservoir and only frequencies near the vicinity of ω_c to be absorbed by the cold reservoir.
- In this model, for every photon of frequency ω_h absorbed from the hot reservoir, a photon of frequency ω_c is emitted into the cold reservoir, and a coherent photon of frequency ν is emitted as work.

The Model

4 Scovil Schulz-DuBois(SSD) Model

- There is a filter that allows only frequencies near the vicinity of ω_h to be absorbed from the hot reservoir and only frequencies near the vicinity of ω_c to be absorbed by the cold reservoir.
- In this model, for every photon of frequency ω_h absorbed from the hot reservoir, a photon of frequency ω_c is emitted into the cold reservoir, and a coherent photon of frequency ν is emitted as work.
- The engine operates in a cycle, where the system absorbs heat from the hot reservoir(allowing transitions with frequency ω_h), performs work(emission with frequency ν), and releases heat(allowing transitions with frequency ω_c) to the cold reservoir.

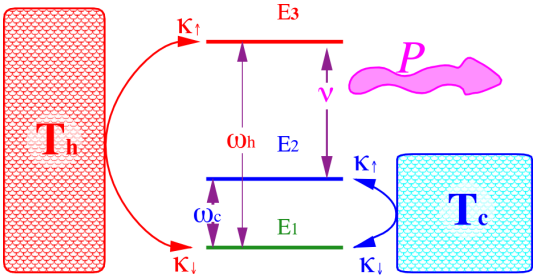

Figure: The Three Level MASER as a Quantum Heat Engine.

Population Distribution

4 Scovil Schulz-DuBois(SSD) Model

- At equilibrium, the populations of the energy levels are given by the Boltzmann distribution:

$$N_i \propto \frac{e^{-E_i/k_B T}}{Z}$$

where k_B is the Boltzmann constant.

Population Distribution

4 Scovil Schulz-DuBois(SSD) Model

- At equilibrium, the populations of the energy levels are given by the Boltzmann distribution:

$$N_i \propto \frac{e^{-E_i/k_B T}}{Z}$$

  where k_B is the Boltzmann constant.

- As a result, the population ratios are:

$$\frac{N_3}{N_1} = e^{-(E_3-E_1)/k_B T_h} = e^{-\hbar\omega_h/k_B T_h}; \frac{N_2}{N_1} = e^{-(E_2-E_1)/k_B T_c} = e^{-\hbar\omega_c/k_B T_c}$$

Population Distribution

4 Scovil Schulz-DuBois(SSD) Model

- At equilibrium, the populations of the energy levels are given by the Boltzmann distribution:

$$N_i \propto \frac{e^{-E_i/k_B T}}{Z}$$

  where k_B is the Boltzmann constant.
- As a result, the population ratios are:

$$\frac{N_3}{N_1} = e^{-(E_3-E_1)/k_B T_h} = e^{-\hbar\omega_h/k_B T_h}; \frac{N_2}{N_1} = e^{-(E_2-E_1)/k_B T_c} = e^{-\hbar\omega_c/k_B T_c}$$

- For stimulated emission to occur, we require a population inversion, which means that the population of the higher energy level must be greater than that of the lower energy level.

Population Inversion and Efficiency

4 Scovil Schulz-DuBois(SSD) Model

Efficiency of the Engine

Similar to a classical heat engine, the efficiency of the SSD engine is given by:

$$\eta_{SSD} = \frac{\text{Work Output}}{\text{Heat Input}} = \frac{\hbar\nu}{\hbar\omega_h}$$

Population Inversion and Efficiency

4 Scovil Schulz-DuBois(SSD) Model

Condition for Population Inversion

The condition for population inversion can be expressed as:

$$e^{-\hbar\omega_h/k_B T_h} > e^{-\hbar\omega_c/k_B T_c}$$

$$\Rightarrow \exp\left(\frac{\hbar\nu}{k_B T_c}\left(\frac{\omega_h(T_h - T_c)}{\nu T_h} - 1\right)\right) > 1$$

One can immediately see that the Carnot efficiency to be defined as $\eta_C = 1 - \frac{T_c}{T_h}$ emerges from this condition. In particular, the condition for population inversion can be rewritten as:

$$\eta_{SSD} \leq \eta_C$$

Engine Efficiency

4 Scovil Schulz-DuBois(SSD) Model

Second Law of Thermodynamics

We can see another verification of the 2nd Law of Thermodynamics here. The efficiency of the engine is given by $\eta_{SSD} = \frac{\nu}{\omega_h}$, and the condition for population inversion is $\eta_{SSD} \leq \eta_C$. This means that the efficiency of the engine can never exceed the Carnot efficiency, which is a fundamental limit set by the second law of thermodynamics.

Summary

4 Scovil Schulz-DuBois(SSD) Model

- An electron in the ground state first absorbs a "hot" photon from the reservoir at T_H temperature, and this takes it to level 3. - Heat intake: $Q_{\text{in}} = E_3$
- The electron undergoes stimulated emission, dropping to level 2 and releasing a coherent photon in the cavity as work. - Work outflow: $W = E_3 - E_2$
- The electron drops from level 2 back to the ground state, dumping a "cold" photon into the bath at temperature T_C. - Heat outflow: $Q_{\text{out}} = E_2$

Table of Contents

5 Modern Work

- Introduction
- MASERs
- Quantum Heat Engines
- Scovil Schulz-DuBois(SSD) Model
- Modern Work
- Conclusion

Quantum Heat Engines

5 Modern Work

- The SSD model was the first example of a quantum heat engine, and it laid the foundation for the field of quantum thermodynamics. Since then, there has been a lot of work done on quantum heat engines, both theoretical and experimental.

Quantum Heat Engines

5 Modern Work

- The SSD model was the first example of a quantum heat engine, and it laid the foundation for the field of quantum thermodynamics. Since then, there has been a lot of work done on quantum heat engines, both theoretical and experimental.
- Quantum Heat Engines can be run in reverse as refrigerators, where work is done to cool of Quantum Computers which are very susceptible to thermal noise.

Quantum Heat Engines

5 Modern Work

- The SSD model was the first example of a quantum heat engine, and it laid the foundation for the field of quantum thermodynamics. Since then, there has been a lot of work done on quantum heat engines, both theoretical and experimental.
- Quantum Heat Engines can be run in reverse as refrigerators, where work is done to cool of Quantum Computers which are very susceptible to thermal noise.
- The SSB Model has been further worked on in the context of the Lindblad Master Equation, which allows for a more rigorous description of the dynamics of the system and the thermodynamic processes involved.

- There have also been more bounds on the engine efficiency , called the Curzon-Ahlborn efficiency, which is given by $\eta_{CA} = 1 - \sqrt{\frac{T_c}{T_h}}$. This efficiency is always less than the Carnot efficiency, and is a more realistic bound for the efficiency of heat engines operating at finite power.
- This limit has been broken in the context of quantum heat engines, where it has been shown that destructive quantum interference effects can allow for efficiencies greater than the Curzon-Ahlborn limit.

Table of Contents

6 Conclusion

- Introduction
- MASERs
- Quantum Heat Engines
- Scovil Schulz-DuBois(SSD) Model
- Modern Work
- Conclusion

Conclusion

- We started by discussing the three-level MASER system, and how it can be used to amplify microwave radiation through the process of stimulated emission.

Conclusion

- We started by discussing the three-level MASER system, and how it can be used to amplify microwave radiation through the process of stimulated emission.
- We then discussed how the three-level MASER system can be interpreted as a quantum heat engine, where the system absorbs heat from a hot reservoir, performs work, and releases heat to a cold reservoir.

Conclusion

- We started by discussing the three-level MASER system, and how it can be used to amplify microwave radiation through the process of stimulated emission.
- We then discussed how the three-level MASER system can be interpreted as a quantum heat engine, where the system absorbs heat from a hot reservoir, performs work, and releases heat to a cold reservoir.
- We also discussed the conditions for population inversion and how they relate to the efficiency of the engine, and how the second law of thermodynamics sets a fundamental limit on the efficiency of quantum heat engines.

Conclusion

- We started by discussing the three-level MASER system, and how it can be used to amplify microwave radiation through the process of stimulated emission.
- We then discussed how the three-level MASER system can be interpreted as a quantum heat engine, where the system absorbs heat from a hot reservoir, performs work, and releases heat to a cold reservoir.
- We also discussed the conditions for population inversion and how they relate to the efficiency of the engine, and how the second law of thermodynamics sets a fundamental limit on the efficiency of quantum heat engines.
- Finally, we briefly touched on some of the modern developments in the field of quantum thermodynamics.

Thank You!

References

Heinz-Peter Breuer and Francesco Petruccione.
The Theory of Open Quantum Systems.
Oxford University Press, Oxford, 2002.

Loris Maria Cangemi, Chitrak Bhadra, and Amikam Levy.
Quantum engines and refrigerators.
2023.

Peter W. Milonni and Joseph H. Eberly.
Laser Physics.
Wiley, Hoboken, NJ, 2010.

H. E. D. Scovil and E. O. Schulz-DuBois.
Three-level masers as heat engines.
Physical Review Letters, 2(6):262–263, March 1959.

Q&A

Thank you for listening!
Your feedback will be highly appreciated!